

Notes: Duffie and Strulovici (Econometrica, 2012)

Capital Mobility and Asset Pricing

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1 Big picture

- **Question.** How do capital-market frictions and financial intermediation shape cross-market risk premia and the speed with which risk premia converge after capital shocks?
- **Core setting.** There are two otherwise identical markets. They differ only in the amount of capital currently invested in each. The market with more capital has a lower risk premium; the market with less capital has a higher risk premium.
- **Main friction.** Investors do not freely and instantly move capital across markets. They move only when contacted by intermediaries, and intermediaries must spend resources to find and move investors.
- **Role of intermediaries.** Intermediaries search for investors in the over-capitalized market and offer to move their capital to the under-capitalized market. They charge a fee equal to a fraction of the investor's present-value gain from switching.
- **Main equilibrium object.** The state variable is the capital imbalance ratio

$$Z_t = \frac{X_t}{Y_t}, \quad X_t = \max\{X_{a,t}, X_{b,t}\}, \quad Y_t = \min\{X_{a,t}, X_{b,t}\}.$$

A high value of Z_t means one market has much more capital than the other.

- **Key theoretical result.** In the baseline homogeneous model, equilibrium intermediation is a trigger rule. Intermediation is zero when markets are not too imbalanced and is at full capacity when imbalance exceeds a threshold T .
- **Asset-pricing implication.** With slow-moving capital, risk premia do not instantly equalize. After a capital shock, the affected market can have a very high risk premium that gradually mean-reverts as intermediaries move capital.
- **Comparative statics.** Capital mobility falls when intermediation costs rise, rises when cross-market payoff differences are larger, falls when investors discount future gains more heavily, and can move ambiguously with asset volatility or search capacity.
- **Competition result.** More intermediaries can increase capital mobility through the standard competition channel, but can also reduce capital mobility by lowering each intermediary's bargaining power and fee share.
- **Important non-neoclassical message.** The model produces persistent return differences across otherwise identical markets without assuming different risk exposures. Return differences reflect the distribution of capital and the speed of intermediation.
- **Bottom line.** The law of one price can fail dynamically because capital is slow to move. Intermediaries help restore pricing parity, but their incentives can make the adjustment incomplete, state-dependent, and sometimes weaker under greater competition.

2 Market environment and key objects

2.1 Agents and markets

The model has two partially segmented markets, labeled a and b . The two markets are symmetric in fundamentals, but can differ in how much capital is currently deployed in each market.

There are three groups of agents:

- **Local hedgers.** They are risk-averse and issue or sell short-lived risky claims inside their local market. They are not mobile across markets.
- **Capital suppliers / investors.** They are risk-neutral investors who provide risk-bearing capital to hedgers. They can move across markets only through intermediation.
- **Intermediaries.** They search for investors and arrange transfers of capital from the low-return over-capitalized market to the high-return under-capitalized market.

Interpretation. The model separates two tasks that are often merged in frictionless asset-pricing models: bearing risk and finding the right market in which to bear that risk. Investors can bear risk, but intermediaries provide the matching and mobility technology.

2.2 Capital levels and dividend rates

Let $X_{a,t}$ and $X_{b,t}$ denote the total capital available in markets a and b . A unit of capital invested in market $i \in \{a, b\}$ earns a dividend or premium rate

$$\pi(X_{i,t}), \tag{1}$$

where $\pi(\cdot)$ is strictly decreasing.

Interpretation. This is the demand side of the model. When a market has abundant capital, hedgers do not need to pay much for risk-bearing capacity, so the risk premium is low. When a market is short of capital, hedgers must pay more, so the risk premium is high.

2.3 Loss processes and risk premia

Capital in each market is exposed to losses. In the baseline formulation, proportional losses arrive according to compound Poisson processes. The common notation is

$$\rho_a = \rho_a^{\text{idiosyncratic}} + \rho^{\text{common}}, \quad \rho_b = \rho_b^{\text{idiosyncratic}} + \rho^{\text{common}}. \tag{2}$$

The two markets are symmetric in the distribution of risks. Therefore, differences in expected returns are not generated by different risk types; they are generated by different capital levels.

If the mean loss arrival rate is η and the mean loss size is $\bar{\nu}$, the net mean return in market i is approximately

$$\pi(X_{i,t}) - \eta\bar{\nu}. \tag{3}$$

Interpretation. The key point is that a market with lower $X_{i,t}$ has a higher $\pi(X_{i,t})$ and therefore a higher expected return, holding risk fixed.

2.4 Frictionless benchmark

If capital were perfectly mobile, investors would immediately move capital to the market with the higher dividend rate. Since $\pi(\cdot)$ is strictly decreasing, this would equalize capital levels:

$$X_{a,t} = X_{b,t} \quad (4)$$

and therefore equalize expected returns.

Interpretation. The paper studies why this frictionless benchmark fails. The failure is not because the assets are fundamentally different; it is because capital does not instantly reach the market where it is most valuable.

2.5 Investor-level capital dynamics

For a particular investor, let C_t be the cumulative net amount of capital moved from market a to market b . The investor's capital in the two markets evolves as

$$dW_{a,t}^C = W_{a,t-}^C d\rho_{a,t} - dC_t, \quad (5)$$

$$dW_{b,t}^C = W_{b,t-}^C d\rho_{b,t} + dC_t. \quad (6)$$

Capital can be moved only when the investor is in contact with an intermediary.

Interpretation. C_t is the investor's reallocation control, but it is not freely chosen at all times. The intermediation friction is a constraint on when capital can actually be moved.

2.6 Intermediation fees

When moving an investor's capital, the intermediary charges a fee equal to a fraction q of the investor's present-value gain from moving. The parameter

$$q \in [0, 1] \quad (7)$$

measures the intermediary's bargaining power.

Interpretation. A higher q means intermediaries keep more of the gains from trade. This makes search more profitable for them, but lowers the investor's net gain from future movement.

2.7 Search technology

An intermediary chooses a contact intensity λ_t . Investors are contacted at the conditional mean rate λ_t . Search has flow cost

$$c\lambda_t, \quad (8)$$

where $c \geq 0$ is the cost coefficient. Search intensity is bounded above:

$$\lambda_t \in [0, \bar{\lambda}]. \quad (9)$$

Interpretation. Intermediation is costly and capacity-constrained. Even when there are gains from moving capital, the intermediary may not search if the gains are too small relative to the cost.

2.8 Aggregate capital flows

Because there is a continuum of investors, the exact law of large numbers implies that aggregate capital flows are deterministic conditional on aggregate states. For example, when market a has more capital than market b , the aggregate rate at which capital moves from a to b is

$$\lambda_t \mathbf{1}_{\{X_{a,t} > X_{b,t}\}} X_{a,t}. \quad (10)$$

Interpretation. A higher contact intensity moves more investors, and a larger over-capitalized market gives the intermediary more capital to move. This is why both search intensity and the capital imbalance matter for adjustment speed.

2.9 State reduction

The two markets are symmetric, so the model can be written in terms of

$$X_t = \max\{X_{a,t}, X_{b,t}\}, \quad Y_t = \min\{X_{a,t}, X_{b,t}\}. \quad (11)$$

The capital imbalance ratio is

$$Z_t = \frac{X_t}{Y_t} \geq 1. \quad (12)$$

Interpretation. Z_t is the central state variable. If $Z_t = 1$, capital is balanced and there is no price pressure. If Z_t is high, one market is capital-scarce and offers a high premium relative to the other.

2.10 Homogeneous payout function

The paper obtains sharp analytical results by using the inverse-demand form

$$\pi(x) = k_0 + \frac{k}{x}, \quad k > 0. \quad (13)$$

The constant k_0 is common across markets and does not affect gains from switching; the cross-market premium difference is driven by the k/x term.

Interpretation. This assumption makes the equilibrium scale-free. What matters is the ratio of capital levels, not the absolute size of both markets.

3 Models and theoretical specifications (all equations + interpretation)

3.1 Capital dynamics under a Markov intermediation policy

Let the intermediation policy be a symmetric Markov function

$$\lambda_t = \Lambda(X_t, Y_t), \quad \Lambda : \mathbb{R}_+^2 \rightarrow [0, \bar{\lambda}]. \quad (14)$$

Under this policy, aggregate capital evolves as

$$dX_t = -\Lambda(X_t, Y_t)X_t dt + X_t d\rho_{X,t}, \quad (15)$$

$$dY_t = \Lambda(X_t, Y_t)X_t dt + Y_t d\rho_{Y,t}. \quad (16)$$

Interpretation. Intermediation drains capital from the over-capitalized market and adds it to the under-capitalized market. Loss shocks move capital in the opposite direction by creating new imbalances.

3.2 Investor value functions

Let $G(x, y)$ be the continuation value per unit of capital when the investor is currently in the over-capitalized market, and let $H(x, y)$ be the corresponding value in the under-capitalized market. The investor prefers H when $x > y$, because the under-capitalized market has a higher premium.

The equations for G and H are

$$rG(x, y) = \pi(x) - G_x(x, y)x\Lambda(x, y) + G_y(x, y)x\Lambda(x, y) \quad (17)$$

$$+ (1 - q)\Lambda(x, y)(H(x, y) - G(x, y)) \quad (18)$$

$$- \eta G(x, y) + \eta(G(x, 0) - G(x, y)), \quad (19)$$

$$rH(x, y) = \pi(y) - H_x(x, y)x\Lambda(x, y) + H_y(x, y)x\Lambda(x, y) \quad (20)$$

$$+ \eta(G(y, 0) - H(x, y)) - \eta H(x, y). \quad (21)$$

Interpretation. These equations are asset-pricing equations for one unit of capital. The value loss from discounting equals dividend income, changes in future state caused by capital flows, expected switching gains net of fees, and expected effects of loss events.

3.3 Switching gain and fee

Given a policy Λ , the gross gain from moving one unit of capital from the over-capitalized market to the under-capitalized market is

$$F^\Lambda(x, y) = H(x, y) - G(x, y). \quad (22)$$

The intermediary's fee per unit moved is

$$qF^\Lambda(x, y). \quad (23)$$

Interpretation. This is the central surplus object. Investors create surplus by moving to the high-premium market, and intermediaries extract fraction q of that surplus.

3.4 Intermediary's optimization problem

If fees are based on a conjectured policy Γ , the monopolistic intermediary's value is

$$V(x, y) = \sup_{\Lambda} \mathbb{E} \left[\int_0^\infty e^{-rt} \Lambda(X_t, Y_t) (X_t q F^\Gamma(X_t, Y_t) - c) dt \right]. \quad (24)$$

An equilibrium is a policy Λ that solves this problem when the conjectured fee-generating policy is the same policy, $\Gamma = \Lambda$.

Interpretation. The intermediary chooses how hard to search. The current benefit of search is the amount of capital that can be moved, X_t , times the fee per unit, qF , minus the flow cost c .

3.5 HJB equation for the intermediary

For a conjectured policy Γ , the HJB equation is

$$0 = \sup_{\delta \in [0, \bar{\lambda}]} \{-rV(x, y) + U(V, x, y, \delta, \Gamma)\}, \quad (25)$$

where

$$U(V, x, y, \delta, \Gamma) = -V_x(x, y)\delta x + V_y(x, y)\delta x \quad (26)$$

$$+ \eta(V(y, 0) + V(x, 0) - 2V(x, y)) \quad (27)$$

$$+ \delta(xqF^\Gamma(x, y) - c). \quad (28)$$

Interpretation. The chosen intensity δ affects current fees and future opportunities. Moving capital today earns fees now but reduces future capital imbalance, which may reduce future fees.

3.6 Bang-bang optimality condition

Because the HJB is linear in the control δ , the optimal search intensity is either zero or maximal. The intermediary searches at full capacity when

$$qF^\Lambda(X_t, Y_t) - (V_x(X_t, Y_t) - V_y(X_t, Y_t)) > \frac{c}{X_t}. \quad (29)$$

Interpretation. The left-hand side is the net marginal benefit of moving one more unit of capital: current fee income minus the loss in future intermediation opportunities. The right-hand side is search cost per unit of movable capital.

3.7 Homogeneity and the switching-gain ODE

With

$$\pi(x) = k_0 + \frac{k}{x}, \quad (30)$$

the equilibrium can be reduced to the capital ratio $z = x/y$. Define

$$f(z) = \frac{F^\Lambda(z, 1)}{k}, \quad L(z) = \Lambda(z, 1). \quad (31)$$

The switching gain satisfies

$$0 = -rf(z) + (1 - z^{-1}) - zL(z)f'(z) + (-f(z) - zf'(z))L(z)z \quad (32)$$

$$- (1 - q)f(z)L(z) + \eta(g_0(1 - z^{-1}) - 2f(z)). \quad (33)$$

Interpretation. This equation tracks the normalized gain from switching. The gain comes from the premium difference $1 - z^{-1}$, is affected by how capital flows change future imbalance, is reduced by fees, and is affected by loss shocks.

3.8 Positive switching gains

The paper proves that for any intermediation policy,

$$f(z) > 0 \quad \text{for all } z > 1. \quad (34)$$

Interpretation. Whenever capital levels differ, an investor in the over-capitalized market strictly wants to switch to the under-capitalized market, provided the opportunity arrives. Therefore, lack of movement comes from the intermediary's incentives and contact frictions, not from investors being indifferent.

3.9 Trigger policy

The equilibrium policy has a trigger form:

$$\Lambda(x, y) = \begin{cases} 0, & x < Ty, \\ \bar{\lambda}, & x \geq Ty, \end{cases} \quad (35)$$

for a unique trigger ratio $T \geq 1$.

Interpretation. Intermediaries do nothing when imbalances are small. Once imbalance is large enough, they search at full capacity. This is the main state-dependent adjustment mechanism of the model.

3.10 Switching-gain equations under the trigger policy

For $z \geq T$, the switching-gain equation becomes

$$(r + 2\eta + \bar{\lambda}[(1 - q) + z])f(z) + \bar{\lambda}(1 + z)zf'(z) = (1 + \eta g_0) \left(1 - \frac{1}{z}\right). \quad (36)$$

For $z \in [1, T]$, because there is no intermediation,

$$(r + 2\eta)f(z) = (1 + \eta g_0) \left(1 - \frac{1}{z}\right). \quad (37)$$

Thus,

$$f(z) = \frac{1 + \eta g_0}{r + 2\eta} \left(1 - \frac{1}{z}\right), \quad z \in [1, T]. \quad (38)$$

Interpretation. Below the trigger, switching is valuable to investors but no intermediary searches. Above the trigger, active intermediation changes the future path of imbalance and therefore changes the value of switching.

3.11 Reduced intermediary value equation

Let

$$v(z) = \frac{V(z, 1)}{k}, \quad v_0 = \frac{V(y, 0)}{k} = \frac{V(x, 0)}{k}. \quad (39)$$

The reduced HJB is

$$0 = \sup_{\delta \in [0, \bar{\lambda}]} \{-rv(z) - \delta z(1+z)v'(z) + 2\eta(v_0 - v(z)) + (qzf(z) - c/k)\delta\}. \quad (40)$$

When $z \in [1, T]$,

$$v(z) = v_1 = \frac{2\eta}{r + 2\eta}v_0. \quad (41)$$

When $z \geq T$,

$$\kappa v(z) + z(1+z)v'(z) = d + qzf(z), \quad (42)$$

where

$$\kappa = \frac{r + 2\eta}{\bar{\lambda}}, \quad d = \frac{2\eta v_0}{\bar{\lambda}} - \frac{c}{k}. \quad (43)$$

Interpretation. The intermediary value is flat below the trigger because no fees are earned before a new shock changes the state. Above the trigger, value rises with imbalance because fee opportunities are larger.

3.12 Smooth pasting and the trigger threshold

The smooth-pasting condition at the trigger is

$$v'(T) = 0. \quad (44)$$

This is equivalent to

$$qTf(T) = \frac{c}{k}. \quad (45)$$

Using the closed-form value of $f(T)$ below the trigger gives

$$T = 1 + \frac{c(r + 2\eta)}{(1 + \eta g_0)qk}. \quad (46)$$

Interpretation. At the trigger, the intermediary is exactly indifferent between searching and not searching. A higher cost c raises T ; a larger fee share q or larger payoff slope k lowers T .

3.13 Conservation equation for g_0

The constant g_0 is pinned down by a conservation equation: the total value of investors plus the intermediary equals the present value of cash dividend payments net of search costs. The paper writes this as

$$kg_0 = \frac{2}{r} - \frac{c\bar{\lambda}}{r} \left(1 - e^{-(2\eta+r)a(T)}\right) - kv_0, \quad (47)$$

where

$$a(T) = \frac{\log(1 + 1/T)}{\bar{\lambda}}. \quad (48)$$

Interpretation. g_0 is not a free parameter. It adjusts so that investor values, intermediary rents, search costs, and dividend payoffs are mutually consistent in equilibrium.

3.14 Existence and uniqueness

The paper proves that there is a unique trigger capitalization ratio T satisfying the switching-gain equations, the smooth-pasting condition, and the conservation equation.

Interpretation. The model delivers a sharp equilibrium prediction: a unique state-dependent law of motion for capital mobility and risk-premium convergence.

3.15 Comparative statics for capital mobility

A higher trigger T means lower capital mobility. In the baseline model:

Parameter increase	Effect on T	Effect on mobility	Intuition
Intermediation cost c	Up	Down	Search is more expensive.
Payout slope k	Down	Up	Cross-market premium gaps are more valuable.
Discount rate r	Up	Down	Investors value future gains less.
Loss intensity η , holding π fixed	Up	Down	Capital imbalance becomes riskier for the intermediary and investors.
Bargaining power $q < 1/2$	Down	Up	Intermediaries keep a larger fee share.
Search capacity $\bar{\lambda}$	Ambiguous	Ambiguous	Higher capacity can speed adjustment but can also lower future rents.

Interpretation. Capital mobility is strongest when gains from moving capital are large and cheap to harvest. It is weakest when search is expensive, agents are impatient, or intermediaries cannot capture enough surplus.

3.16 Oligopolistic intermediaries with fixed bargaining power

Suppose there are n identical intermediaries, each with capacity $\bar{\lambda}/n$, so aggregate market capacity remains $\bar{\lambda}$. With fixed q , the reduced HJB for an intermediary is

$$0 = \sup_{\delta \in [0, \bar{\lambda}/n]} \left\{ -rv(z) - \left(\frac{n-1}{n} \bar{\lambda} \mathbf{1}_{\{z \geq T\}} + \delta \right) zv'(z) \right. \quad (49)$$

$$\left. - \left(\frac{n-1}{n} \bar{\lambda} z \mathbf{1}_{\{z \geq T\}} + \delta z \right) zv'(z) + 2\eta(v_0 - v(z)) \right. \quad (50)$$

$$\left. + (qzf(z) - c/k)\delta \right\}. \quad (51)$$

The same smooth-pasting condition applies:

$$qTf(T) = \frac{c}{k}. \quad (52)$$

Interpretation. With fixed bargaining power and full-loss shocks, splitting a fixed aggregate search capacity across more intermediaries does not change the trigger. Each intermediary's private problem leads to the same aggregate trigger rule.

3.17 Endogenous bargaining power

The paper then allows bargaining power to depend on market structure. If a fraction ψ_n of investors are well-connected and can bargain with more than one intermediary, the average intermediary fee share is

$$q(n) = 0 \cdot \psi_n + \frac{1}{2}(1 - \psi_n) = \frac{1 - \psi_n}{2}. \quad (53)$$

If ψ_n rises with n , then $q(n)$ falls with the number of intermediaries.

Interpretation. Competition can make investors harder to extract rents from. This lowers the fee share of each intermediary and can weaken search incentives.

3.18 Simple-strategy equilibrium under endogenous bargaining

Let $L_i(z)$ be the search intensity of intermediary i , and let aggregate search be

$$L(z) = \sum_{i=1}^n L_i(z). \quad (54)$$

For simple strategies, each $L_i(z)$ is either 0 or $\bar{\lambda}/n$. If exactly n_j intermediaries are active on an interval of capital ratios, the equilibrium inequalities are

$$q(n_j)zf^L(z) - \frac{c}{k} \geq 0, \quad (55)$$

$$q(n_j + 1)zf^L(z) - \frac{c}{k} \leq 0. \quad (56)$$

Interpretation. Active intermediaries must find search profitable, while the next inactive intermediary must find entry unprofitable. As imbalance rises, more intermediaries may become active.

4 Equilibrium and comparison logic

4.1 What identifies risk-premium differences in the model

Risk-premium differences are pinned down by capital levels, not by differences in fundamental risk. Since both markets are symmetric in loss distributions, any difference in expected returns comes from the strictly decreasing payout function $\pi(X)$.

Interpretation. The model isolates a pure capital-imbalance channel. This makes the asset-pricing mechanism clean: unequal capital produces unequal risk premia.

4.2 Why price adjustment is gradual

After a capital shock, the affected market becomes under-capitalized and offers a higher premium. Capital does not instantly flow in because investors need contact with intermediaries. Intermediaries search only when doing so is privately profitable.

Interpretation. The speed of return convergence is an equilibrium object, not an exogenous adjustment parameter. It depends on search costs, fee shares, loss risk, discounting, and market structure.

4.3 Why the model is not just a transaction-cost model

A pure transaction-cost model would generally imply an immediate price adjustment to a new wedge. In this paper, the key empirical signature is different: prices or risk premia jump after a shock and then mean-revert as capital is gradually redeployed.

Interpretation. The model is designed to explain delayed adjustment, not merely a static no-trade band.

4.4 Why Markov equilibrium matters

The intermediary cannot commit to future search intensity. Therefore, the relevant equilibrium is Markov: at each state (X, Y) , the intermediary chooses the search intensity that is optimal given current and future incentives.

Interpretation. Lack of commitment prevents the intermediary from promising future policies that would be optimal only from an ex ante perspective. This makes the trigger policy a true dynamic equilibrium object.

4.5 Why the trigger rule is credible

The HJB is linear in the search control. Therefore, whenever search is worthwhile, it is optimal at the maximum feasible rate; whenever it is not worthwhile, it is optimal to do none. This produces the bang-bang policy.

Interpretation. The trigger rule is not imposed for convenience. It follows from the linear search technology and bounded intensity.

4.6 Why homogeneity is powerful

The assumption $\pi(x) = k_0 + k/x$ lets the model reduce from two capital levels (x, y) to one ratio $z = x/y$. This makes the switching gain, intermediary value, and threshold functions of z alone.

Interpretation. Homogeneity is what makes the analytical characterization possible. It turns a potentially high-dimensional dynamic problem into a one-dimensional trigger problem.

4.7 How competition has two opposing effects

Competition has two conceptually distinct channels:

- **Business-stealing / non-internalization channel.** Each intermediary does not fully internalize how its search reduces future opportunities for other intermediaries. This can increase aggregate search.
- **Bargaining-power channel.** More competition can make investors better connected and reduce the fee share $q(n)$. This can decrease aggregate search.

Interpretation. The paper's nonstandard result is that competition need not always raise capital mobility. If competition mainly lowers intermediary rents, it can reduce the incentive to search for movable capital.

4.8 What the model can and cannot distinguish

The model can explain why otherwise similar assets have persistent expected-return differences after shocks to capital supply. It cannot by itself identify whether observed price reversals are caused by search frictions, information-processing frictions, institutional constraints, or other slow-moving-capital mechanisms.

Interpretation. The paper gives a disciplined mechanism and comparative statics. Empirical work still needs to separate this mechanism from other sources of delayed price adjustment.

4.9 Empirical settings the paper has in mind

The discussion section points to several empirical contexts where capital appears slow-moving:

- catastrophe reinsurance after large loss events,
- investor inattention and infrequent portfolio rebalancing,
- post-news price momentum,
- equity supply shocks and index recompositions,
- corporate bond downgrades and defaults,
- convertible bond and credit-market dislocations.

Interpretation. These settings share the same time signature: an immediate price or premium response followed by gradual reversal as capital is located and redeployed.

5 Main propositions and theoretical implications

The paper is theoretical. Its main “exhibits” are propositions and equilibrium equations rather than empirical tables or figures.

5.1 Proposition 1: HJB verification

Read: If a bounded differentiable function solves the intermediary's HJB equation, then it is the value function, and any measurable policy attaining the supremum is optimal.

Economic message. The intermediary's dynamic search problem is well-posed. The HJB approach is not just heuristic; it characterizes the value and optimal policy.

5.2 Proposition 2: investors strictly gain from switching when capital is imbalanced

Read: For any intermediation policy Λ , the normalized switching gain satisfies

$$f(z) > 0 \quad \text{for } z > 1. \quad (57)$$

Economic message. Investors in the over-capitalized market want to move to the under-capitalized market. The friction is not a lack of gains from trade; it is the cost and incentive problem of finding and moving capital.

5.3 Proposition 3: intermediary value is increasing in imbalance

Read: For any trigger ratio T , the solution $v(z)$ to the reduced HJB is bounded and increasing, and is strictly increasing on $[T, \infty)$.

Economic message. Greater capital imbalance creates more valuable intermediation opportunities. This is why search effort increases with imbalance.

5.4 Propositions 4 and 5: unique trigger equilibrium

Read: There exists a unique trigger ratio T . In equilibrium,

$$\lambda_t = 0 \quad \text{if } Z_t < T, \quad \lambda_t = \bar{\lambda} \quad \text{if } Z_t \geq T. \quad (58)$$

If search cost is zero, then $T = 1$ and the intermediary always works at full capacity.

Economic message. The model gives a sharp state-dependent law of one-price adjustment. The economy behaves nearly frictionlessly only when intermediation is cheap and capacity is high.

5.5 Proposition 6: comparative statics for costs, payoffs, discounting, and loss risk

Read: Capital mobility is decreasing in the intermediation cost c , increasing in the payout coefficient k , decreasing in the discount rate r , and decreasing in the loss intensity η when the dividend function π is held fixed.

Economic message. Markets with high search costs, impatient agents, or less valuable cross-market premium gaps should show slower convergence of risk premia after shocks.

5.6 Proposition 7: bargaining power

Read: For $0 < q < 1/2$, capital mobility is increasing in the intermediary's bargaining power q .

Economic message. The more surplus intermediaries can capture, the more they search. This creates a tradeoff: high intermediary rents can be socially costly, but they may also speed capital reallocation.

5.7 Proposition 8: oligopoly with fixed bargaining power

Read: With n intermediaries and fixed bargaining power, there is a unique symmetric Markov homogeneous equilibrium. In the full-loss benchmark, the trigger ratio equals the trigger of a monopolist with the same aggregate maximum contact intensity $\bar{\lambda}$.

Economic message. Simply splitting the same aggregate search capacity across more intermediaries does not necessarily change mobility if bargaining power is fixed. Competition matters more once it affects rents or post-shock incentives.

5.8 Proposition 9: more active intermediaries at larger imbalances

Read: If $q(n)$ is decreasing in n , then the number of active intermediaries is increasing in the capital imbalance ratio.

Economic message. When markets are only mildly imbalanced, few intermediaries may find it profitable to search. When imbalances are large, more intermediaries enter because gross gains are larger.

5.9 Propositions 10 and 11: competition can reduce mobility

Read: For sufficiently infrequent loss events, the reduction in bargaining power caused by oligopolistic competition can make the region of full capital mobility smaller than under monopoly. In this sense, more intermediaries can reduce capital mobility.

Economic message. Competition is not automatically pro-mobility. If competition erodes the fee share that motivates search, intermediaries may invest less in finding investors, and capital can remain misallocated for longer.

5.10 Discussion section: asset-pricing implications

Read: The paper emphasizes several implications:

- capital shocks generate larger immediate risk-premium responses than in a perfect-mobility model;
- risk premia mean-revert as capital is redeployed;
- risk-premium dispersion is higher when capital is more unevenly distributed;
- adjustment speed depends on intermediation costs, bargaining power, discounting, volatility, and market structure.

Economic message. Expected returns can be high not because assets are fundamentally riskier, but because the right capital has not yet arrived.

5.11 Evidence discussed in the paper

Read: The paper links the theory to evidence from reinsurance, portfolio inattention, slow post-news price adjustment, equity supply shocks, index recompositions, and credit-market dislocations.

Economic message. These examples all display the same dynamic pattern: an initial price-pressure effect followed by gradual reversal. That is the empirical signature of slow-moving capital.

6 Short summary

- The paper builds a dynamic model of two segmented but otherwise identical asset markets.
- The market with more capital has a lower risk premium; the market with less capital has a higher risk premium.
- Investors can move capital only through intermediaries who must pay search costs.
- Intermediaries charge fees equal to a share of investors' gains from switching.
- In the baseline model, equilibrium intermediation is a unique trigger policy: no search below T , full search above T .
- Higher search costs and higher discount rates reduce capital mobility.
- Larger payoff gaps and higher intermediary bargaining power raise capital mobility.
- Search capacity and asset volatility can have ambiguous effects.
- More intermediaries can increase mobility through competition, but can also reduce mobility by lowering fee shares and search incentives.
- The model explains persistent cross-market expected-return differences and gradual return convergence after capital shocks.

7 Conclusion

7.1 Core conclusion

Duffie and Strulovici show that asset-pricing deviations across otherwise identical markets can be generated by slow-moving capital. When capital is unevenly distributed, the capital-scarce market offers a higher risk premium. Intermediaries move capital toward that market, but only when the private value of search exceeds the cost.

7.2 Economic intuition

The economic mechanism is simple but powerful:

$$\text{capital imbalance} \Rightarrow \text{return gap} \Rightarrow \text{intermediation surplus} \Rightarrow \text{gradual capital movement.} \quad (59)$$

Because the final step is costly and incentive-constrained, return gaps can persist and then mean-revert over time.

7.3 Takeaway

The paper turns the phrase “limits to arbitrage” into an explicit dynamic equilibrium model of search, fees, and capital flows. The most important lesson is that intermediaries are not passive conduits. Their search costs, bargaining power, and competitive environment determine how quickly capital reaches high-return opportunities and how quickly the law of one price is restored.